Report

INTER DEPARTMENTAL PROJECT-III

on

DESIGN OF HYBRID COUPLER IN RF PASSIVE COMPONENTS

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(ACCREDITED BY **NAAC** WITH **"A+"** GRADE)

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CERTIFICATE

This is to certify that the Minor project/IDP report entitled **"DESIGN OF HYBRID COUPLER IN RF PASSIVE COMPONENTS"** that is being submitted by students B. Abhignu (201FA05003), K. Manaswini(201FA05058), B. Rahul(211LA05009), and A. Sairakesh (211LA05032) to the Department of Electronics and Communication Engineering, Vignan's Foundation for Science Technology and Research, is a record of work carried out by them under the guidance of Dr.K.Venkata Kishore,Ph.D, department of Electronics and Communication Engineering.

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ABSTRACT

Due to the rapid development of wireless communication technologies, the number of wireless users are radically increasing. Currently, wireless devices are connected to the internet, and these numbers are expected to increase manifolds in the years to come.

The technology growth of the fifthgeneration (5G) wireless systems will be needed to meet this high demand of the network. 5G wireless systems offer data-rates of up to 10Gbps, 1-ms latency, and reduced power consumption. It is a known fact that 5G wireless systems will be exploiting beyond the presently used 3 GHz microwave and millimetrewave (mm-wave) frequency bands.

In current 5G and future Beyond 5G (B5G) antenna array systems, beamforming networks (BFNs) such as the Butler Matrix (BM) will play a key role in achieving multi-beam characteristics. The BMs are classified into the bi-layer, tri-layer, and four-layer structures. It includes different techniques that have been used to solve the problem of crossing, narrow bandwidth, and size reduction of the BM

OBJECTIVE

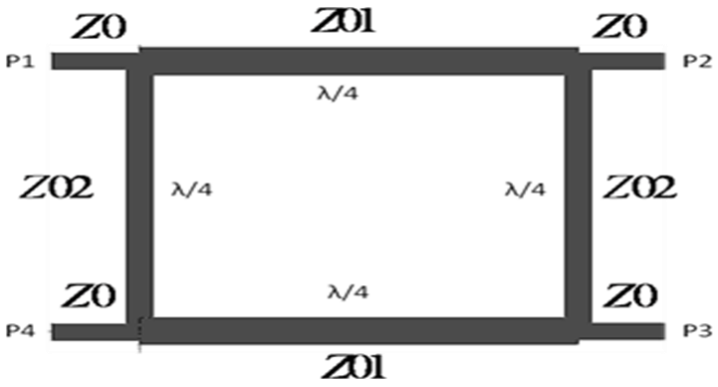
- The primary objective of DESIGN OF HYBRID COUPLER IN RF PASSIVE COMPONENTS. The components that can store energy in the form of voltage or current .example: resister, capacitors and inductors
- The RF passive component that covers Circulators, Isolators, Power Dividers, Couplers, RF connectors, RF wires, RF Cable assemblies, Loads, Attenuators, SAW devices etc.
- The **directional couplers** are passive devices used mostly in the field of radio technology.
- An essential feature of directional couplers is that they only couple power flowing in one direction
- Hybrid couplers are important components in RF systems that allow for power splitting and phase shifting
- Design considerations, such as power handling, frequency range, and impedance matching, are important when selecting and designing hybrid couplers.

INTRODUCTION

- In an antenna array each element must be fed with proper amplitude and phase such that the array will perform desired functions. The feeder network that distributes the proper amplitude and phase excitation to each element . The microwave power to the radiating antenna is feeding through a coaxial cable like RG214. the feeder network consist of coaxial cables , power dividers/combiners and hybrid couplers with equal/unequal amplitude 90 degree phase difference.
- Design and analysis of unequal power butler matrix feeding network(2:2) using hybrid coupler for low SLL x-band radar applications.
- Design and analysis of equal power butler matrix feeding network(2:2) using hybrid coupler for low SLL x-band radar radar applications.
- Design of butler matrix feeding network with the help of hybrid couplers.

FEEDING NETWORK MODEL(HYBRID COUPLER)

The hybrid coupler (quadrature coupler) is a well-known circuit in microstrip technique used for its ability to generate signals 90 degree out of phase at its output. It consists, as shown in Figure , of a main line that is coupled to a secondary line by two quarter-wavelength long sections spaced one quarter wavelength apart, thus creating a square approximately one wavelength in circumference



When a signal is introduced at port-1, the signals at the two output arms (2,3) are equal in amplitude but have a 90degrees phase difference between each other .

The port located on the same side as the input port is isolated since there is no power reaching it.

Z01 and Z02 denote the characteristic impedances of the series and shunt lines respectively.

The power applied at a port is evenly distributed between the ports located on the opposite side of the coupler.

There is a 90 degree phase difference between these two ports, the port closer to the input port is leading in phase by 90 degree .

The port located on the same side as the input port is isolated since there is no power reaching it. In the same way by varying the impedances of the opposite arms in a branch line coupler, unequal power splits can be obtained.

SOFTWARE REQUIRED

- Advanced Design System (ADS) is an electronic design automation software system produced by Path Wave Design, a division of Keysight Technologies. It provides an integrated design environment to the designers of RF electronics products such as mobile phones, pagers, wireless network, satellite communications, radar systems and high speed data links.



WORKING PRINCIPLE

DESIGN OF HYBRID COUPLER WITH EQUAL OUTPUTS:

For designing of hybrid coupler(equal power splitter) we are considering the frequency range of 9.5GHZ and the power handling of this coupler is Z0=50 ohms and Z0a=Z0/ √ 2=50/ √ 2=35.35 ohms .

Here Z0a=Z0/√2.

Here Z0a splits the power in equal in equal ratio across output ports (i.e across port 2 and 3).

STEP-2 :

For calculating the width and height of the coupler

W/H = 2/π[B − 1 − ln(2B − 1) + εr-1/2εr[ln(B-1)+0.39-(0.61/ εr)]]

initialized all the values ,then W/H = 3.080

and W= 2.464mm

STEP-3 :

For finding the effective relative permeability

Ereff = εr+1/2+ εr-1/2[1+12H/W]^(-1/2)

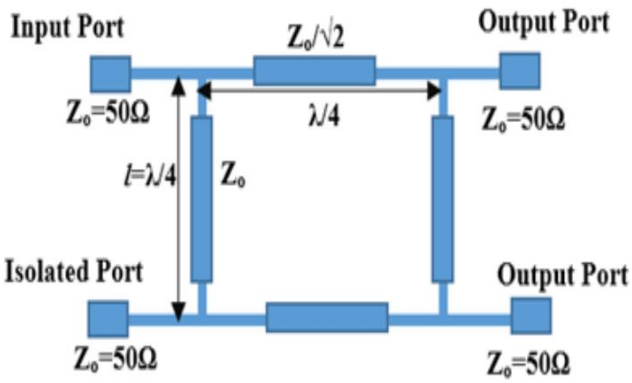
Ereff=1.87

STEP-4:

We use to find the guided wave length

$\lambda g = \lambda 0/\sqrt{\text{Ereff}}$

λg =23.09



STEP- 5 :

For finding the length of the wave guide(L)

$$L = \frac{\lambda g}{4}$$

L = 5.77mm

Now for finding the Z0a values the same procedure can be taken as previous caculations

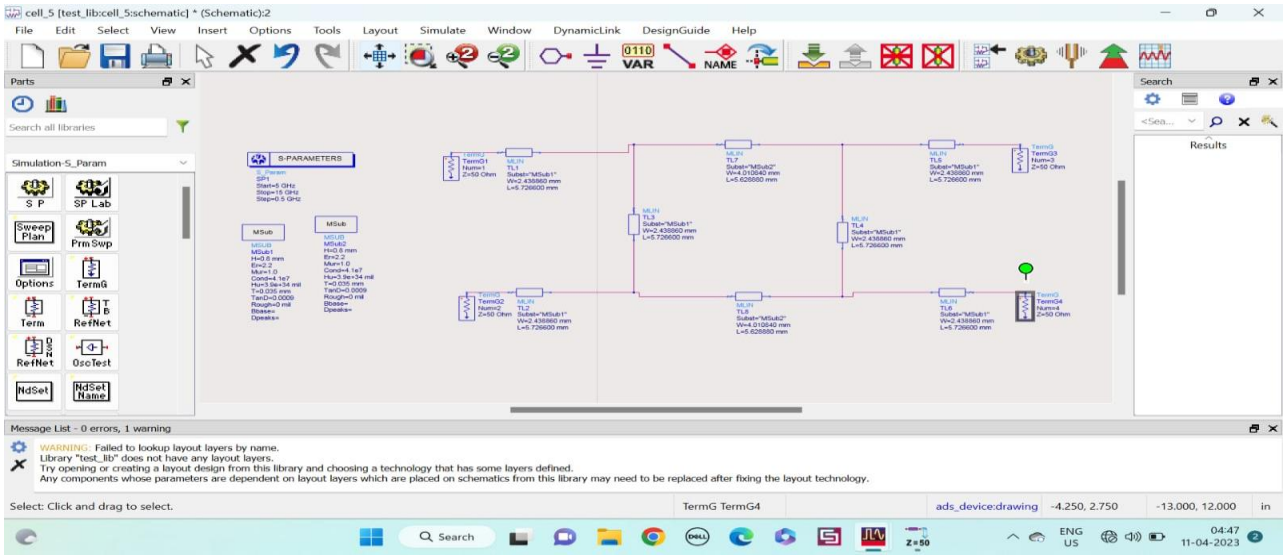
B=11.29

W=3.98mm

Ereff=1.924

$\lambda g = 22.76$ and L = 5.69mm

DESIGN OF EQUAL AMPLITUDE POWER SPLITTER



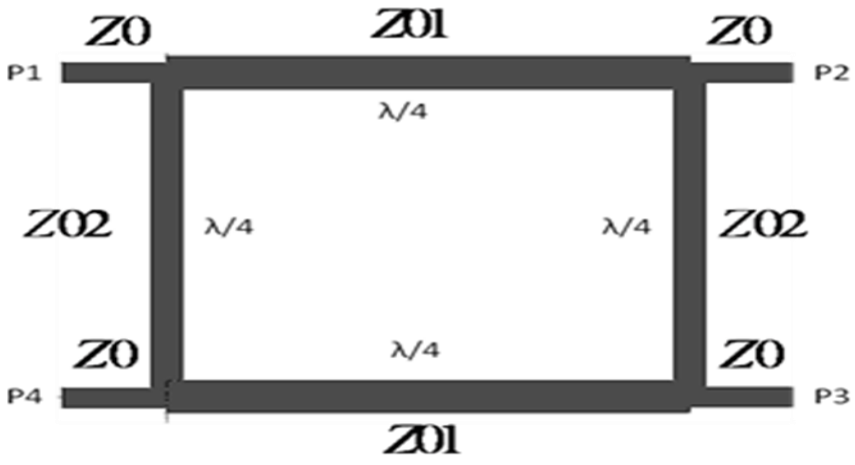
DESIGN OF UNEQUAL AMPLITUDE POWER SPLITTER

- For designing of hybrid coupler(equal power splitter) we are considering the frequency range of 9.5GHZ and the power handling of this coupler is Z0=50 ohms
- Here Z01 and Z02 splits the power in equal in unequal ratio across output ports (i.e across port 2 and 3).Let the powers across port1 and port2 are 0.6 and 0.4 respectively , with the input power applied across port1 is 1.
- The advantages of this type of unequal power splitter reduces the side lobe levels in the radiation pattern hence increase in the gain and directivity .Main lobe level increases , maximum power is concentrated at main lobe level results in increase of directivity and gain properties.

CALCULATIONS OF Z01 & Z02:

- Z01=Z0 √ [(PA/PB)/(1+(PA/PB))]
- Z02= Z0 √ [(PA/PB)] , where Z0=50 ohms
- Here PA=0.4 and PB=0.4 , as mentioned in previous slide, by substituting values of PA & PB and calculating the values of Z01 & Z02
- Z01=Z0 √ [(0.6/0.4)/(1+(0.6/0.4))]
    - Z01=50 √ (1.5/2.5)
    - Z01=50(0.7745)=38.725
    - Z02= Z0 √ [(0.6/0.4)]
    - Z02= 50 √(1.5)=61.234

Hence Z01=38.725 & Z02=61.234



The remaining design procedure follows the same steps as in equal power splitter:

1.B=377 π/2Z0 √εr

2. W/H = 2/π[B − 1 − ln(2B − 1) + εr-1/2εr[ln(B-1)+0.39-(0.61/ εr)]]

3. Ereff = εr+1/2+ εr-1/2[1+12H/W]^(-1/2)

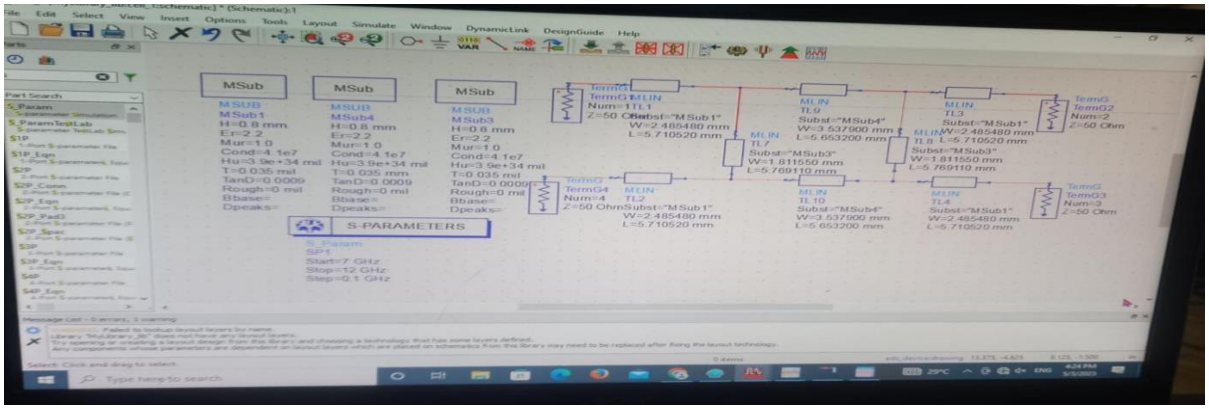
4. $\lambda g = \lambda 0/\sqrt{Ereff}$

5. L = $\frac{\lambda g}{4}$,

- where H=0.8mm, εr=2.2, t=0.035, frequency=9.5G Hz
- By computing the above five steps for Z0=50 , Z01=38.725 , Z02=61.235
- We get,

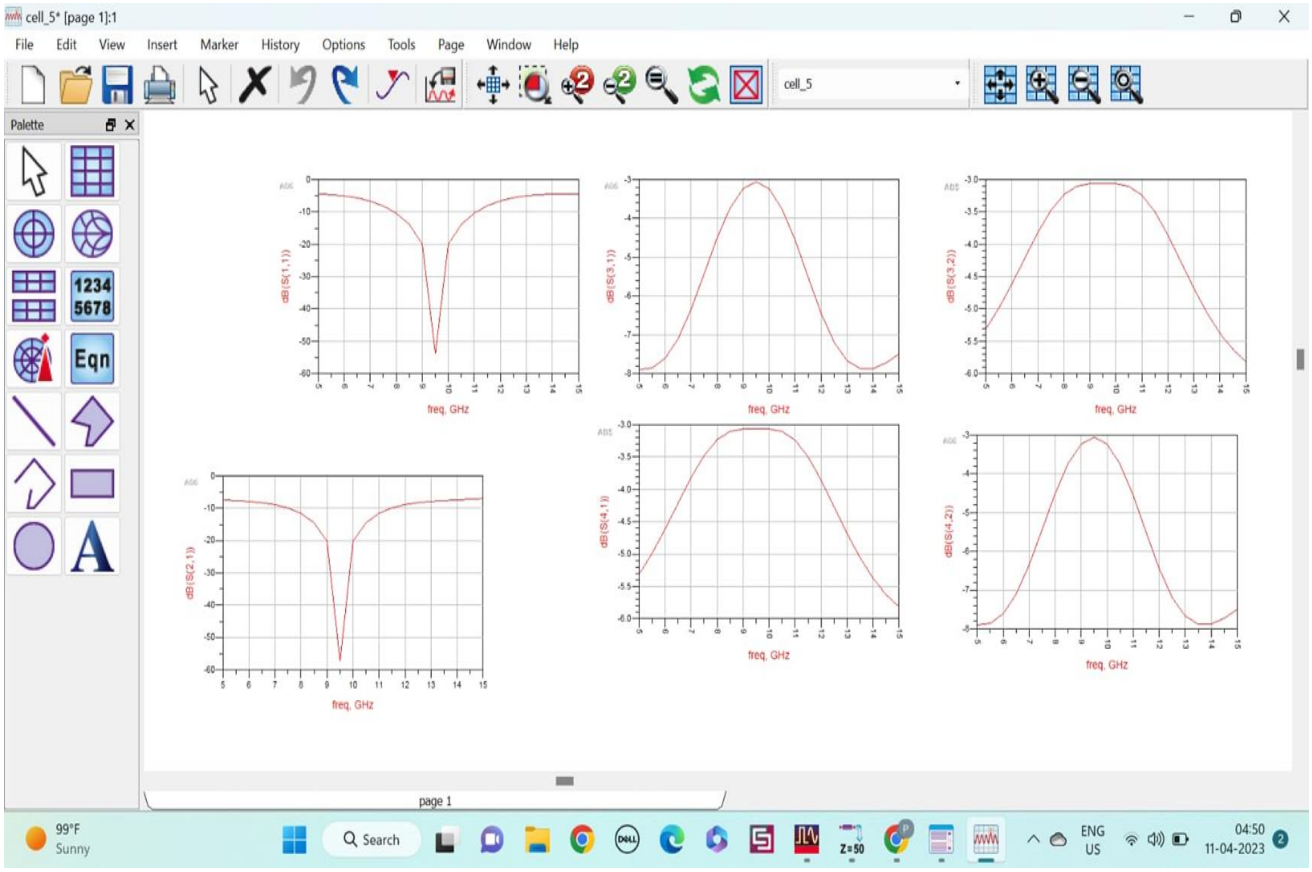
PARAMETER	IMPEDANCE(ohms)	LENGTH(MM)	WIDTH(MM)
Z0	50	5.7214	2.4649
Z01	38.725	5.6714	4.4527
Z02	61.235	5.8229	1.7969

DESIGN OF UNEQUAL AMPLITUDE POWER SPLITTER

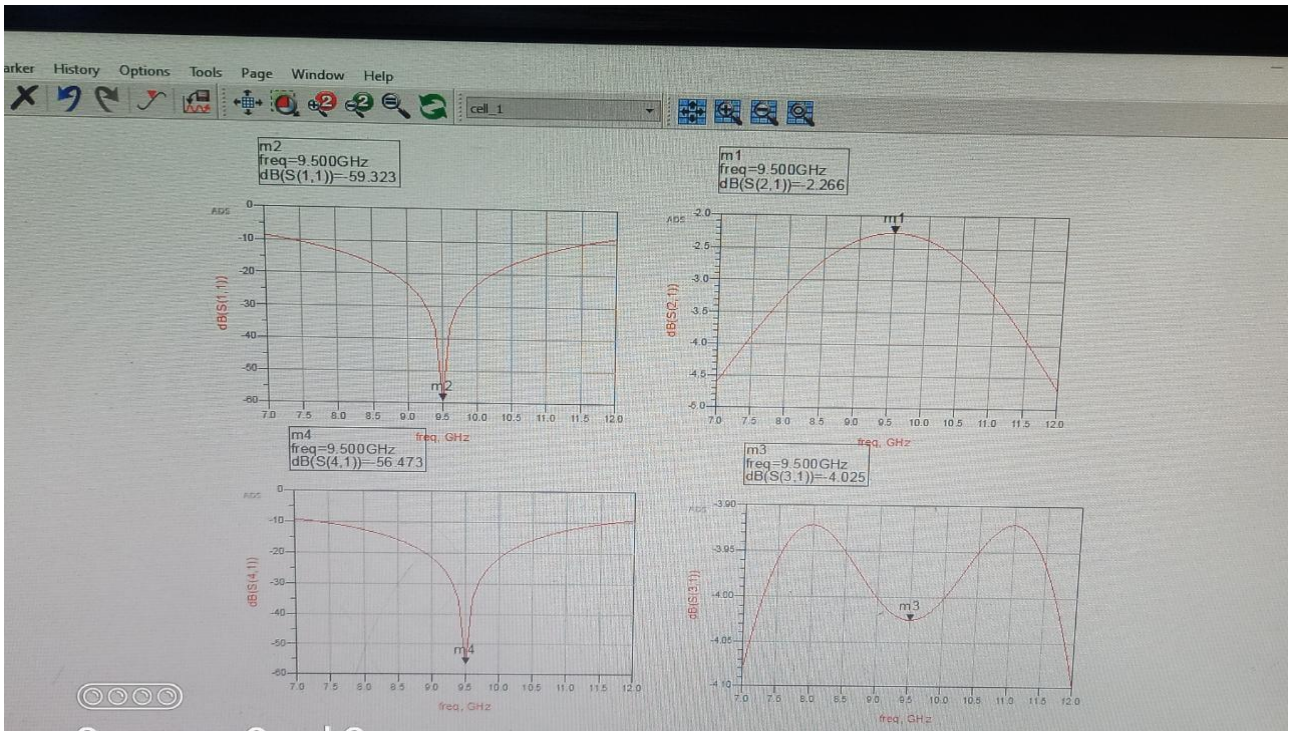


OUTPUT

S PARAMETERS: FOR EQUAL



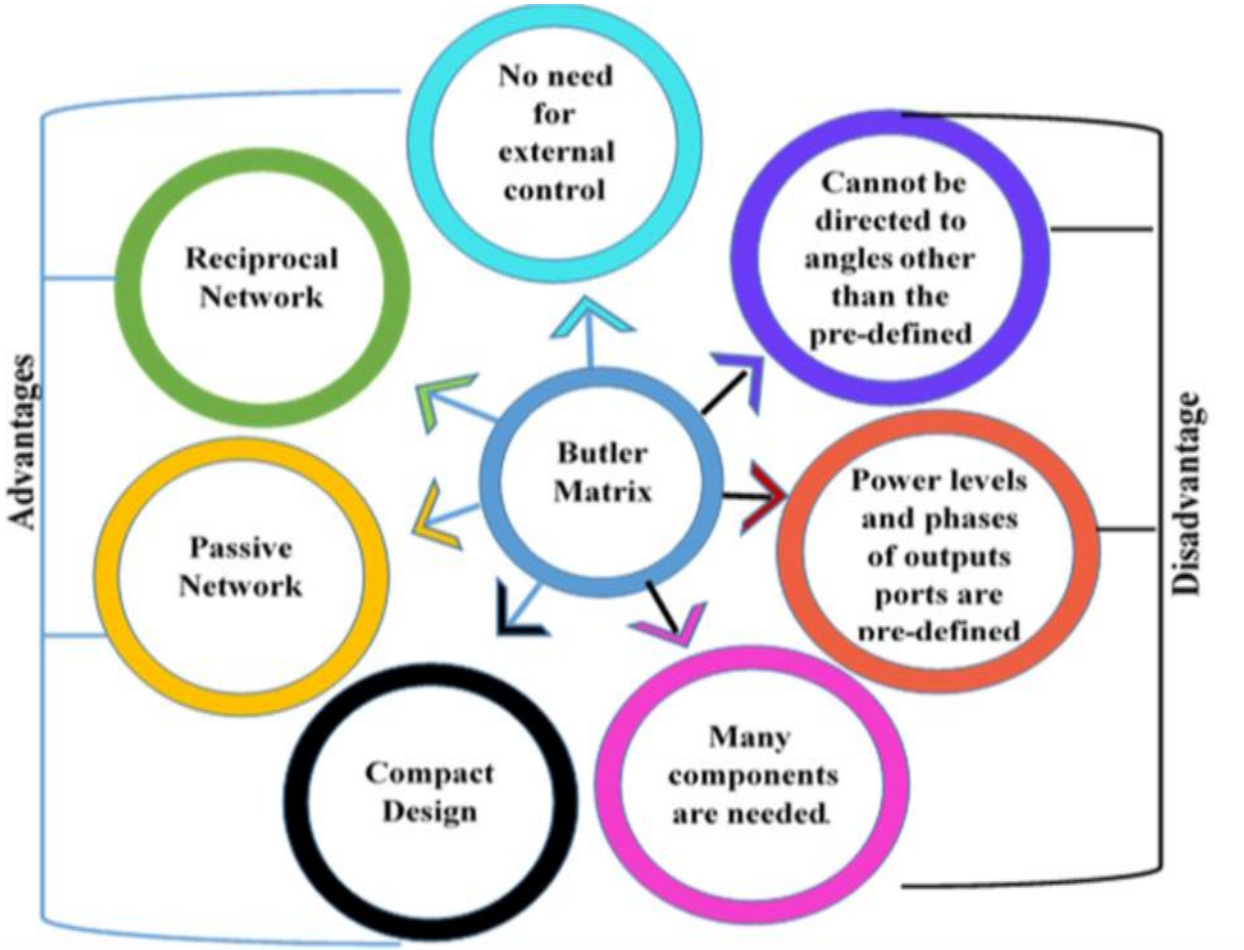
S PARAMETERS: FOR UNEQUAL



IMPLEMENTATION OF HYBRID COUPLER

- Cellular Base Stations: Used to split power between multiple antennas
- Radar Systems: Used to split power between multiple receivers
- Satellite Communications: Used to split power between multiple transmitters

ADVANTAGES AND DISADVANTAGES



FUTURE SCOPE OF THE PROJECT

BUTTLER MATRIX :

- The BM is a BFN that is used to feed the multiple antenna array element. The BM has '*N*' input ports connected to the main transceiver to get a signal and '*N*' output ports connected to the antenna elements of the phase array antenna (PAA) system. With a progressive phase difference between the output ports, the BM feeds the signal from the transceiver to the antenna elements so that the beamforming operation can be performed, and the signal can be transmitted in different directions.
- A *N*×*N* BM is a passive beamforming network(BFN) consisting of *(N/2)*log2N* branch-line coupler (BLCs), *N/2*log2(N-1)* phase-shifters, and 0dB crossovers
- For 2x2 BM, which consists of two 90o BLCs for designing of hybrid coupler
- At the output of the 2x2 BM it consists of two antenna which are connected to each other**.**

CONCLUSION

- Hybrid couplers are important components in RF systems that allow for power splitting and phase shifting.
- There are different types of hybrid couplers, each with their own unique characteristics and applications.
- Design considerations, such as power handling, frequency range, and impedance matching, are important when selecting and designing hybrid couplers.
- Performance optimization techniques, such as tuning and impedance matching, can be used to improve the performance of hybrid couplers.
- Careful consideration of tradeoffs between performance parameters is necessary when selecting and designing hybrid couplers.

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